Faddeev–Amplituhedron–MZM Research Program

A Four-Stage Mathematical Physics Research Project Plan

Principal Investigator: Robert | **Institution:** [Research Institution]

Document Date: 26 May 2026 | **Version:** 1.0 | **Classification:** Research Planning — Confidential Draft

Program Duration: 6 Months (Sprint-Based) | **Funding Target:** NSF / DOE Mathematical Physics Program

Keywords: Faddeev quantum dilogarithm, amplituhedron, Majorana zero modes, categorification, QAOA, QSVT, knot theory

---

Table of Contents

Executive Summary

Research Vision

This program proposes a unified, four-stage mathematical physics research initiative establishing the **Faddeev quantum dilogarithm pentagon identity** as the organizing principle connecting knot topology, higher-categorical algebra, quaternionic amplituhedron geometry, and topological quantum computation. The research proceeds from rigorous analytic foundations through to prototype quantum hardware experiments over a six-month sprint-based timeline.

The Faddeev–Amplituhedron–MZM (FAM) Research Program addresses one of the most compelling open problems at the intersection of mathematical physics and quantum information science: whether the deep combinatorial and algebraic structures underlying scattering amplitudes in quantum field theory — epitomized by the amplituhedron — are connected, via quantum dilogarithm identities, to the topological degrees of freedom that protect quantum information in Majorana zero mode (MZM) architectures.

The program's **unifying hypothesis** is that the Faddeev quantum dilogarithm functional equation, understood as a pentagon identity in a suitable monoidal category, simultaneously governs: (a) q-series invariants of hyperbolic 3-manifolds (knot complements), (b) wall-crossing phenomena in Donaldson-Thomas theory of Calabi-Yau 4-folds, (c) the positive Grassmannian cell decomposition of a quaternionic amplituhedron in 8 real dimensions, and (d) locality constraints in QAOA-style quantum optimization encoding MZM logical qubits.

Stage 1 (Foundational) formalizes the Faddeev quantum dilogarithm $\Phi_b(x)$ over non-commutative matrix rings and quaternion modules, computing state integrals $Z(M; q)$ for the figure-eight (4_1) and 5_2 knots and validating against published q-series tables. **Stage 2 (Algebraic Lift)** categorifies the pentagon identity as a coherence 2-morphism in an A_∞ category and defines twisted Neumann-Zagier matrices in a dg-algebra setting, mapping to Donaldson-Thomas partition functions. **Stage 3 (Geometric Model)** constructs an 8-manifold quaternionic amplituhedron from BCFW-style tetrahedral triangulations, sampling its geometry via RJMCMC and applying the framework to Kagome/moiré BEC vortex lattice simulations. **Stage 4 (Algorithmic)** encodes the resulting structures into GQPSO-CCEA hybrid optimizers and QAOA circuits with NZ-matrix-derived locality constraints, prototyping MZM qubit encodings via QSVT block-encoding on IBM Quantum hardware.

Expected outcomes include: (1) a peer-reviewed paper on quaternionic extensions of the Faddeev dilogarithm; (2) open-source Python/Julia software libraries covering all four stages; (3) the first computational demonstration of an 8-dimensional quaternionic amplituhedron via RJMCMC; (4) a prototype QAOA circuit benchmarked on 50–127 qubit IBM Quantum devices; and (5) a proof-of-concept MZM qubit encoding with simulated fidelity ≥ 0.99. The total computational budget spans approximately 7,700 CPU-hours, 2,400 GPU-hours, and 500 QPU-hours across six months, requiring no novel hardware procurement beyond cloud access.

★ Program Novelty Statement

No prior work has unified the Faddeev pentagon identity, quaternionic amplituhedron geometry, and Majorana zero mode encoding within a single computational research program. This program constitutes the first systematic attempt to establish this triangle of connections, with falsifiable predictions at each stage verifiable against established mathematical benchmarks.

---

1. Program Overview

1.1 Vision Statement

The FAM Research Program is founded on the conviction that the deepest structures of contemporary mathematical physics — quantum topology, higher algebra, positive geometry, and topological quantum computation — are unified by a single organizing algebraic identity: the **pentagon identity of the Faddeev quantum dilogarithm**. This identity, first identified by Faddeev in the context of quantum groups and integrable systems (1994), has since appeared independently in knot theory, cluster algebra wall-crossing, scattering amplitude geometry, and — we conjecture — in the locality structure of quantum error-correcting codes built on Majorana zero modes.

Our vision is to make this unity *explicit, computable, and experimentally testable* through a rigorously staged research program combining algebraic geometry, category theory, high-performance scientific computing, and hybrid classical-quantum optimization. The program is designed to produce immediate mathematical results (Stages 1–2) that directly feed into computational geometry experiments (Stage 3) and quantum algorithm prototypes (Stage 4), with each stage providing validation inputs for the next.

1.2 Unifying Hypothesis

π Core Hypothesis: Pentagon Identity as Universal Organizing Principle

The Faddeev quantum dilogarithm pentagon identity

$$\Phi_b(p)\,\Phi_b(q) = \Phi_b(q)\,\Phi_b(p+q)\,\Phi_b(p) \quad \text{for } [p,q] = 1/2\pi i$$

simultaneously governs: knot complement q-series (Stage 1), dg-categorical wall-crossing in DT theory (Stage 2), BCFW triangulation of a quaternionic amplituhedron (Stage 3), and QAOA locality constraints encoding MZM qubits (Stage 4). A single deformation parameter $b \in \mathbb{C}$ threads through all four manifestations.

1.3 Four-Stage Architecture

The program is organized into four sequential but overlapping stages, each building on the mathematical and computational infrastructure of its predecessors:

Stage	Title	Mathematical Goal	Computational Mapping	Physical / Experimental Target
1	Foundational	Formalize Faddeev dilog on matrix/quaternion modules	Symbolic algebra + numeric state-integral evaluation	Validate on simple knot complements (4_1, 5_2)
2	Algebraic Lift	Categorify pentagon; define twisted NZ matrices in higher algebra	Implement algebraic identities in tensor network code	Map to 4D topological indices (DT, GV)
3	Geometric Model	Build 8-manifold amplituhedron from triangulation data	Discretize for Monte Carlo / RJMCMC sampling	Simulate Kagome/moiré BEC vortex lattices
4	Algorithmic	Encode into GQPSO-CCEA / QAOA locality constraints	Hybrid classical-quantum optimizer; block-encoding QSVT	Prototype qubit scaling and MZM encodings

1.4 Program-Level Success Criteria

- At least one preprint submitted per stage upon completion of that stage's milestones.
- All software released open-source under MIT license with DOI-citable Zenodo deposits.
- End-to-end pipeline integration test passing: Stage 1 knot data → Stage 4 qubit measurement → recovered knot invariant.
- All Stage 4 QPU experiments reproducible across at least three independent runs.

---

2. Stage 1 — Foundational: Faddeev Dilogarithm on Matrix/Quaternion Modules

2.1 Scope and Objectives

Stage 1 establishes the rigorous analytic and computational foundation for the entire program. The central task is to generalize the Faddeev quantum dilogarithm $\Phi_b(x)$ — classically defined for a single commuting variable x — to the non-commutative setting of matrix rings $M_n(\mathbb{C})$ and quaternion modules $\mathbb{H}$. This generalization is non-trivial because the functional equations of Φ_b rely crucially on the Baker-Campbell-Hausdorff formula, which receives corrections in non-commutative settings. We must also define and numerically evaluate module-valued state integrals Z(M; q) for hyperbolic knot complements triangulated by SnapPy.

Primary Objectives:

1. Derive and verify a quaternionic extension of the Faddeev functional equation $\Phi_b(x)\Phi_b(-x) = e^{\pi i x^2}\Phi_b(0)^2$, accounting for $\mathbb{H}$-valued operator ordering.
2. Construct the matrix-valued R-matrix from ideal triangulation data of hyperbolic knot complements via Neumann-Zagier (NZ) matrices.
3. Compute and verify q-series factorizations $Z(\lambda, \tau) = \Sigma f_\lambda(q)g_\lambda(\tilde{q})$ for the 4_1 (figure-eight) and 5_2 knots, matching published tables to order q^{20}.

2.2 Mathematical Definitions

∇ Key Definition 1.1: Faddeev Quantum Dilogarithm

The Faddeev quantum dilogarithm is defined for $\mathrm{Im}(b^2) > 0$ by the convergent integral:

$$\Phi_b(x) = \exp\left(\int_{\mathbb{R}+i0} e^{2\pi x t} / (\sinh(\pi b t)\sinh(\pi t/b)) \cdot dt/(4t) \right)$$

where $b \in \mathbb{C}$ is the deformation parameter with $\mathrm{Re}(b) > 0$, and $x \in \mathbb{R}$ (extended to a strip in $\mathbb{C}$ by analytic continuation). The function satisfies $\Phi_b(x + ib/2) = (1 + e^{2\pi b x})\Phi_b(x - ib/2)$.

∇ Key Definition 1.2: Pentagon / Star-Triangle Identity

For operators p, q satisfying the Heisenberg relation $[p, q] = 1/(2\pi i)$, the pentagon identity reads:

$$\Phi_b(p)\, \Phi_b(q) = \Phi_b(q)\, \Phi_b(p+q)\, \Phi_b(p)$$

This identity plays the role of the Yang-Baxter equation for the quantum dilogarithm and is the algebraic backbone of all four stages of the program.

∇ Key Definition 1.3: Matrix-Valued State Integral

For a triangulated hyperbolic 3-manifold M with ideal triangulation τ, the matrix-valued state integral is:

$$Z_M(\tau) = \int \prod_{t \in \tau} \Phi_b(z_t) \cdot \delta(\text{gluing constraints})\, dz_1 \ldots dz_N$$

where $z_t \in \mathbb{H}$ are quaternion-valued shape parameters of tetrahedra and the gluing constraints encode the NZ matrix symplectic data (A, B) with $A^T B - B^T A = 0 \bmod 2$.

∇ Key Definition 1.4: q-Series Factorization Theorem

The state integral admits a q-series factorization (Garoufalidis-Kashaev, 2013):

$$Z(\lambda, \tau) = \Sigma_\lambda\, f_\lambda(q)\, g_\lambda(\tilde{q}) \quad \text{with } q = e^{2\pi i\tau},\ \tilde{q} = e^{-2\pi i/\tau}$$

where $f_\lambda(q)$ and $g_\lambda(\tilde{q})$ are holomorphic and anti-holomorphic q-series, respectively, labeled by "color" parameter λ in a discrete spectrum determined by the NZ data.

Neumann-Zagier (NZ) Matrices: For each ideal triangulation of a cusped hyperbolic 3-manifold, SnapPy computes a symplectic pair (A, B) of integer matrices encoding the holonomy gluing equations. The symplecticity condition $A^T B - B^T A \equiv 0$ (mod 2) is a necessary consistency condition that our code verifies as a unit test. The quaternionic extension replaces the commuting shape parameters $z_t \in \mathbb{C}$ with $\mathbb{H}$-valued operators satisfying modified gluing relations.

2.3 Computational Experiments

Software Environment

Package	Version	Purpose
`Python`	3.11	Primary scripting

Package	Version	Purpose
		language
`SymPy`	≥ 1.13	Non-commutative symbolic algebra; pentagon identity derivation
`SageMath`	≥ 10.3	Algebraic topology; Clifford/quaternion algebra packages
`mpmath`	≥ 1.3	Arbitrary-precision complex integration (state integral)
`numba`	≥ 0.59	JIT compilation of inner integration loops
`numpy / scipy`	≥ 1.26 / 1.13	Numerical arrays, residue computation
`snappy`	≥ 3.2	SnapPy knot complement triangulation data (JSON export)
`pytest`	≥ 8.0	Automated unit test suite

Experiment 1: Symbolic Quaternionic Pentagon Identity

Using SymPy's `NonCommutativeSymbol` algebra, define quaternion-valued operators P, Q satisfying $[P, Q] = 1/(2\pi i) \cdot \mathbf{1}_{\mathbb{H}}$, where $\mathbf{1}_{\mathbb{H}}$ is the identity quaternion. Expand Φ_b as a formal power series in P, Q to degree 8, and symbolically verify both sides of the pentagon identity agree term-by-term. Record the first-order quaternionic correction term (the 'twist') arising from $ij = k \neq ji$.

Experiment 2: Numerical State Integral for 4_1 Knot

Load the ideal triangulation of the figure-eight knot complement $M = S^3\backslash 4_1$ from SnapPy (2 ideal tetrahedra, shape parameter $z \approx 0.5 + 0.866i$). Evaluate $Z_M(b)$ numerically for $b = 1 + 0.1i$ using `mpmath.quad` with working precision 50 decimal digits. Compare to Garoufalidis-Kashaev (2013) Table 1 values to 12 decimal places.

Experiment 3: q-Series Extraction via Residue Computation for Figure-Eight Knot

Using the Cauchy residue method applied to the state integral after contour deformation, extract the q-series coefficients $f_\lambda(q)$ for $\lambda = 0, 1, \ldots, 10$. Represent M as a 2-tetrahedra SnapPy triangulation exported to JSON (see data format below). Verify factorization $Z(\lambda, \tau) = f_\lambda(q)g_\lambda(\tilde{q})$ holds to order q^{20}.

Data Format Specification

📄 SnapPy Triangulation JSON Schema

Triangulation files exported from SnapPy as `knot_triangulations/M41.json` with schema:

{ "manifold": "4_1", "num_tetrahedra": 2, "tetrahedra": [{ "tet_index": 0, "gluing_matrix": [[0,1,0,0],[1,0,0,0],[0,0,0,1],[0,0,1,0]], "shape_parameter": {"re": 0.5, "im": 0.8660254037844387} }, ...], "NZ_A_matrix": [[1,0],[0,1]], "NZ_B_matrix": [[0,1],[1,0]] }

The primary deliverable notebook is `Stage1_faddeev.ipynb` with clearly labeled cells for each experiment, narrative markdown, and embedded unit test calls.

2.4 Validation Tests

- **q-Series Benchmark:** Reproduce known q-series coefficients for 4_1 and 5_2 knots (Garoufalidis-Kashaev 2013, Table 1) to order q^{20}. All coefficients must match the published integer values exactly.
- **Quaternionic Functional Equation:** Verify $\Phi_b(x)\Phi_b(-x) = e^{\pi i x^2}\Phi_b(0)^2$ numerically for 5 random quaternion-valued x and 3 values of b, to 12 decimal places using `mpmath` at 50-digit precision.
- **NZ Symplecticity:** Automatically verify $A^T B - B^T A \equiv 0 \pmod 2$ for every triangulation loaded from SnapPy. Test against the full SnapPy cusped manifold census (1,267 manifolds).

- **Pentagon Unit Tests:** `pytest` suite verifying the scalar pentagon identity $\Phi_b(p)\Phi_b(q) = \Phi_b(q)\Phi_b(p+q)\Phi_b(p)$ for 10 random b values sampled from the strip $\mathrm{Re}(b) \in (0.5, 1.5)$, $\mathrm{Im}(b) \in (-0.2, 0.2)$, tolerance 10^{-10}.

2.5 Resource Estimates

Resource	Specification	Estimated Consumption	Notes
CPU	16-core workstation (e.g., AMD Ryzen 9 7950X)	~200 CPU-hours	Symbolic sweeps + numeric q-series
RAM	32 GB DDR5	Up to 28 GB peak	Large q-series coefficient arrays
Storage	Local SSD	~10 GB	SnapPy triangulation databases
GPU	Not required	—	All computation is CPU/symbolic
Quantum hardware	Not required	—	No quantum resources at Stage 1
Software	SnapPy, SageMath, Python ecosystem	Zero cost	All open-source

2.6 Milestones and Deliverables

Milestone	Target Date	Deliverable	Acceptance Criterion
M1.1	Month 1, Week 2	NZ matrix code validated on SnapPy cusped manifold library; `nz_matrices.py` module	All 1,267 manifolds pass symplecticity check; test runtime < 60 s
M1.2	Month 2, Week 2	Quaternionic	Symbolic proof

Milestone	Target Date	Deliverable	Acceptance Criterion
		pentagon identity proved symbolically (SymPy proof certificate) and numerically verified	complete to degree 8; numeric check to 12 decimal places for 5 test quaternions
M1.3	Month 3, Week 2	State integral code for 4_1 and 5_2 knots; reproducible Jupyter notebook `Stage1_faddeev.ipynb`	q-series coefficients match published tables to order q^{20}; notebook runs end-to-end in < 2 hours on specified hardware

2.7 Risk Register

Risk	Likelihood	Impact	Mitigation Strategy
Quaternionic non-commutativity breaks standard residue calculus; contour integration ill-defined for ℍ-valued integrand	M	H	Use Clifford algebra package in SageMath; rewrite integrals in $sl_2(\mathbb{H})$ matrix representation; consult Andersen-Kashaev (2014) framework; fallback: restrict to complex 2×2 matrix representation of ℍ
SnapPy triangulation data format changes between versions, breaking JSON import	L	M	Pin SnapPy to version 3.2 in `environment.yml`; archive raw .tri files in `data/knot_triangulations/`; write format-agnostic

Risk	Likelihood	Impact	Mitigation Strategy
			parser
State integral fails to converge for complex b with |Im(b)| > 0.3 due to oscillatory integrand	M	M	Analytic continuation via Cauchy contour deformation; fallback to Padé approximants from convergent subseries; increase `mpmath` working precision to 100 digits

2.8 Collaborators and Literature

Key References: Faddeev (1994, quantum dilogarithm and quantum groups); Garoufalidis–Kashaev (2013, arXiv:1304.2705, q-series for knot complements); Andersen–Kashaev (2014, TQFT from quantum Teichmüller theory); Galakhov–Mironov–Morozov (2015, arXiv:1510.05366, pentagon identities and DT); Zagier (2007, "The dilogarithm function" in *Frontiers in Number Theory, Physics, and Geometry II*).

Suggested Collaborators: Stavros Garoufalidis (Georgia Tech / MPIM Bonn) — quantum topology and q-series; Kazuhiro Hikami (Kyushu / IPMU Tokyo) — quantum Teichmüller and modular forms; Jørgen Ellegaard Andersen (Aarhus / SDU) — TQFT and quantum dilogarithm.

---

3. Stage 2 — Algebraic Lift: Categorification and Twisted NZ Matrices

3.1 Scope and Objectives

Stage 2 elevates the scalar dilogarithm identities of Stage 1 to the richer language of higher algebra. The primary mathematical task is *categorification* of the pentagon identity: replacing the scalar equation with a natural transformation between functors on a 2-category, so that the identity holds "up to coherent homotopy" in the sense of Boardman-Vogt-May operads. Simultaneously, we define *twisted Neumann-Zagier matrices* $A_\theta = A + \theta B$ ($\theta \in \mathbb{H}$) in a dg-algebra (differential graded algebra) or $A\infty$-algebra setting, and map the resulting categorical invariants to 4D topological indices: Donaldson-Thomas (DT) and Gopakumar-Vafa (GV) partition functions for Calabi-Yau fourfolds.

Primary Objectives:

4. Define the categorified pentagon identity as a coherence 2-morphism η: $\Phi_b(-) \circ \Phi_b(-) \Longrightarrow \Phi_b(-) \circ \Phi_b(-+-) \circ \Phi_b(-)$ in a Fukaya-type $A\infty$ category.
5. Construct the dg-algebra of twisted NZ matrices with differential $d_A = [A_\theta, \cdot]$ and compute its cohomology for test knot complements.
6. Match DT partition function coefficients for at least 3 Calabi-Yau test geometries (local $\mathbb{P}^2$, local $\mathbb{P}^1 \times \mathbb{P}^1$, and the conifold).

3.2 Mathematical Definitions

∇ Key Definition 2.1: Categorified State Complex

Replace the scalar state integral $Z(M; q)$ with a cochain complex $C^*(M; \mathcal{L})$ with objects in a dg-category $\mathscr{D}$:

$$C^*(M; \mathcal{L}) = (\oplus_p C^p(M; \mathcal{L}), d_{NZ})$$

where the differential d_{NZ} encodes the NZ gluing data as a Maurer-Cartan element in the dg-Lie algebra of the triangulation. The Euler characteristic recovers $\chi(C^*) = Z(M; q)|_{q=1}$.

∇ Key Definition 2.2: Twisted NZ Matrices in dg-Algebra

For deformation parameter $\theta \in \mathbb{H}$, the twisted NZ matrix is:

$$A_\theta = A + \theta B \in M_n(\mathbb{H})$$

with twisted differential $d_{A\theta} = [A_\theta, \cdot]$ acting on the dg-algebra $\Omega^*(M) \otimes M_n(\mathbb{H})$. Symplecticity is deformed: $A_\theta^\dagger B - B^\dagger A_\theta = 2\theta(B^\dagger B) \bmod 2$.

∇ Key Definition 2.3: Donaldson-Thomas Partition Function

For a Calabi-Yau threefold X (our target: fourfold X_4 via dimensional reduction), the DT partition function is:

$$Z_{DT}(X; q) = \Sigma_{n\geq 0}\, \chi(\mathrm{Hilb}^n(X)) \cdot q^n = M(-q)^{\chi(X)} \cdot (\text{quantum corrections})$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function and $\chi(\mathrm{Hilb}^n(X))$ are topological Euler characteristics of Hilbert schemes. Pentagon coherence constraints predict specific wall-crossing jumps in Z_{DT} across stability chambers.

Pentagon Natural Transformation: In the language of A_∞ categories (Fukaya, 2002; Kontsevich-Soibelman, 2008), the pentagon identity lifts to a coherence 2-morphism satisfying the MacLane pentagon coherence axiom for monoidal functors. This is the algebraic heart of Stage 2: the pentagon is not merely a numerical identity but a structural constraint on the entire categorical framework.

3.3 Computational Experiments

Software Environment

Package	Platform	Purpose
`Macaulay2`	v1.24 (standalone)	dg-algebra cohomology; Gröbner basis for twisted NZ ideal
`SageMath`	≥ 10.3	Quiver representations; Ext/Tor computations
`ITensor.jl`	Julia 1.10	MPO (Matrix Product Operator) for pentagon coherence
`quimb`	Python ≥ 1.8	Tensor network contraction (fallback to Python if Julia unavailable)
`networkx`	Python ≥ 3.3	Quiver diagram construction and analysis
`Mathematica`	v14 (optional)	Symbolic tensor computation cross-check

Experiment 1: Cohomology of Twisted NZ Complex

For the trefoil (3_1) and figure-eight (4_1) knot complements, construct the twisted NZ chain complex $C^*(M; A_\theta)$ in Macaulay2 for $\theta \in \{0, i, j, k, 1+i\}$. Compute $H^*(C^*(M; A_\theta))$ using Macaulay2's `HH` functor. Verify that H^0 recovers the classical Alexander polynomial evaluation and that $\chi(C^*) = Z(M; q=1)$ as computed in Stage 1.

Experiment 2: Pentagon Coherence in Tensor Network Language

Represent the pentagon identity as a PEPS (Projected Entangled Pair States) tensor network contraction using ITensor.jl with bond dimension $D = 16$. Implement five-index tensors $T[i,j,k,l,m]$ encoding the 2-morphism data. Verify MacLane coherence (the pentagon axiom for monoidal categories) by checking that two distinct contraction orderings of the 5-tensor network give numerically identical results, to tolerance 10^{-8}, for 5 random quaternionic inputs.

Experiment 3: DT Partition Function Matching

For local $\mathbb{P}^2$ (KK3-fibered), local $\mathbb{P}^1\times\mathbb{P}^1$, and the resolved conifold, compute the DT partition function $Z_{DT}(q)$ to order q^{20} using the MacMahon function formula with quantum corrections from our twisted NZ data. Compare coefficient by coefficient against OEIS sequence A008897 (MacMahon plane partitions) and known geometric results from Maulik-Nekrasov-Okounkov-Pandharipande (2006).

3.4 Validation Tests

- **Euler Characteristic:** Verify $\chi(C^*(M; A_\theta))$ equals the classical NZ determinant det(A) for $\theta = 0$ on all test manifolds (5 manifolds from Stage 1).
- **Pentagon Coherence:** Check MacLane coherence theorem numerically for 5 test A_∞ categories constructed from random NZ data; all pentagon diagrams must commute to tolerance 10^{-8}.
- **DT Coefficients:** Coefficients of $Z_{DT}(q)$ match OEIS sequence A008897 for the MacMahon piece, and match Maulik et al. (2006) Proposition 4.2 for the quantum correction piece, to order q^{10}.
- **Twisted Symplecticity:** Verify the deformed symplecticity relation $A_\theta^\dagger B - B^\dagger A_\theta = 2\theta(B^\dagger B) \bmod 2$ numerically for all 5 quaternion values of θ.

3.5 Resource Estimates

Resource	Specification	Estimated Consumption	Notes
CPU	32-core cluster node (e.g., AMD EPYC 7543)	~500 CPU-hours	Tensor contraction sweeps (bond dim D=16–64)
RAM	50 GB per node	Up to 48 GB peak	ITensor.jl MPO state storage
Storage	Cluster scratch	~50 GB	Tensor network checkpoints
GPU	Not required	—	Classical tensor

Resource	Specification	Estimated Consumption	Notes
			network is CPU-bound
Quantum simulator	Not required	—	Classical TN sufficient for bond dim ≤ 64

3.6 Milestones and Deliverables

Milestone	Target Date	Deliverable	Acceptance Criterion
M2.1	Month 3, Week 2	Twisted NZ matrix definition and dg-algebra structure coded in Macaulay2; `twisted_nz.py` Python interface	Cohomology computed for trefoil and figure-eight; Euler characteristic matches det(A) for θ=0
M2.2	Month 4, Week 2	Pentagon categorification as 2-morphism; `tensor_network.py` MPO implementation	MacLane pentagon coherence verified for all 5 test A_∞ categories; tensor equality to 10^{-8}
M2.3	Month 5, Week 2	DT index matching for local $\mathbb{P}^2$, local $\mathbb{P}^1\times\mathbb{P}^1$, and conifold; `Stage2_algebraic.ipynb`	DT coefficients match to genus 5 (q^{10}); pentagon coherence verified for all associativity constraints in test set

3.7 Risk Register

Risk	Likelihood	Impact	Mitigation Strategy
A_∞ coherence	**M**	**M**	Use operadic

Risk	Likelihood	Impact	Mitigation Strategy
verification intractable symbolically for large operadic compositions			approach via Koszul duality (Loday-Vallette 2012); reformulate as homotopy-coherent diagrams in Lean 4 for machine-verified proofs; fallback: verify only up to 3-fold composition
ITensor MPO memory explosion for large bond dimension D ≥ 64	**M**	**M**	Cap bond dimension at D = 64 with DMRG-style truncation; use SVD truncation with discarded weight < 10^{-10}; switch to quimb (Python) if Julia memory overhead too large
DT partition function quantum correction terms diverge for non-compact geometries	**L**	**H**	Use Nekrasov Ω-background regularization; restrict to compact local models first; consult Okounkov's vertex formalism

3.8 Collaborators and Literature

Key References: Kontsevich–Soibelman (2008, wall-crossing formula and motivic DT); Bridgeland (2010, stability conditions on triangulated categories); Khovanov–Lauda (2009, categorification of quantum groups); Huang–Zhang–Ye (2024, arXiv:2405.19077, pentagon equations in 4D topological orders); Maulik–Nekrasov–Okounkov–Pandharipande (2006, DT/GW correspondence).

Suggested Collaborators: Maxim Kontsevich (IHES) — wall-crossing and motivic DT; Tom Bridgeland (Sheffield) — stability conditions; Andrei Okounkov (Columbia) — vertex formalism and DT theory.

4. Stage 3 — Geometric Model: 8-Manifold Amplituhedron from Tetrahedral Triangulations

4.1 Scope and Objectives

Stage 3 is the geometric heart of the program. We construct a *quaternionic/octonionic generalization* of the Arkani-Hamed–Trnka amplituhedron as an 8-dimensional geometric object arising from the same BCFW-style tetrahedral triangulation data as the knot complements in Stage 1. The classical (real) amplituhedron $A_{n,k,m}$ lives inside the real Grassmannian Gr(k, k+m); our quaternionic amplituhedron $A_{n,k,4}^{\mathbb{H}}$ lives inside the quaternionic Grassmannian $Gr_{\mathbb{H}}(k, k+4) \subset \mathbb{H}P^{n-1}$, which as a real manifold is 8-dimensional for k=1. We discretize its geometry for Monte Carlo and RJMCMC sampling, and apply the resulting statistical framework to Kagome/moiré Bose-Einstein condensate (BEC) vortex lattice simulations.

Primary Objectives:

7. Enumerate BCFW-style triangulation cells $C_\alpha \subset Gr_{\mathbb{H}}(k, n)$ for (n, k) = (8, 2) and verify cell count formula C(n−4, k) from classical theory.
8. Implement and validate an RJMCMC sampler on the quaternionic amplituhedron volume form, achieving acceptance rates in the target range [0.15, 0.45].
9. Simulate Kagome/moiré vortex lattice configurations on a triangulated 2-torus T^2, reproducing the Berezinskii-Kosterlitz-Thouless (BKT) transition signature in the topological charge distribution.

4.2 Mathematical Definitions

∇ Key Definition 3.1: Standard (Real) Amplituhedron

The amplituhedron $A_{n,k,m}$ is the image of the positive Grassmannian $Gr_{\geq 0}(k, n)$ under the map:

$$\tilde{Z}: Gr_{\geq 0}(k, n) \to Gr(k, k+m), \quad C \mapsto C \cdot Z$$

for a matrix $Z \in Mat_{n\times(k+m)}^{>0}$ with all maximal minors positive. The BCFW triangulation gives $A_{n,k,m} = \cup_\alpha C_\alpha$ with $C(n-k-m, k)$ cells (for m=4, $C(n-4-k, k)$ cells).

∇ Key Definition 3.2: Quaternionic Amplituhedron

Replace $\mathbb{R}$ with $\mathbb{H}$ in the amplituhedron construction. The quaternionic Grassmannian $Gr_{\mathbb{H}}(k, n)$ parametrizes k-dimensional right $\mathbb{H}$-subspaces of $\mathbb{H}^n$. The quaternionic amplituhedron is:

$$A_{n,k,4}^{\mathbb{H}} = \tilde{Z}_{\mathbb{H}}(Gr_{\mathbb{H},\geq 0}(k, n)) \subset Gr_{\mathbb{H}}(k, k+4)$$

As a real manifold, $Gr_{\mathbb{H}}(1, 2) \cong S^4$ (quaternionic projective line, $\dim_{\mathbb{R}} = 4k(n-k)$ for $Gr_{\mathbb{H}}(k,n)$), placing our primary target $A_{8,2,4}^{\mathbb{H}}$ in a 24-real-dimensional ambient space with an 8-real-dimensional distinguished stratum.

∇ Key Definition 3.3: RJMCMC Sampler on Cell Decomposition

Reversible-jump MCMC (RJMCMC; Green, 1995) samples across cell decompositions of varying dimension via transdimensional jump proposals. For our amplituhedron:

- **Within-cell moves:** Standard Metropolis-Hastings on BCFW cell C_α with log-probability proportional to $\log|\Omega_\alpha|$ (amplituhedron volume form)
- **Jump proposals:** Cell splitting ($C_\alpha \to C_{\alpha 1} \cup C_{\alpha 2}$) and cell merging, with Jacobian correction ensuring detailed balance

- **Target distribution:** $\int_A \Omega \propto$ tree-level scattering amplitude for given (n, k)

∇ Key Definition 3.4: BEC Vortex Lattice on Triangulated Torus

The Kagome/moiré magnetic unit cell is discretized as a triangulated 2-torus T^2 with:

- Vortex positions = 0-simplices of triangulation (36 sites for 6×6 unit cells, 3 Kagome sites each = 108 vortices)
- Topological charge = winding number of phase field around each plaquette: $q_p = (1/2\pi) \oint_{\partial p} \nabla\theta \cdot dl \in \mathbb{Z}$
- BKT transition: divergence of topological susceptibility $\chi_{top} = \langle(\Sigma\, q_p)^2\rangle - \langle\Sigma\, q_p\rangle^2$ at temperature T_{BKT}

4.3 Computational Experiments

Software Environment

Package	Platform	Purpose
`NumPy / SciPy`	Python 3.11	Grassmannian coordinate arrays, triangulation geometry
`PyMC`	≥ 5.10 (Python)	RJMCMC framework with JAX backend
`ArviZ`	≥ 0.19	MCMC diagnostics ($\hat{R}$, ESS, autocorrelation)
`JAX`	≥ 0.4.25	GPU-accelerated log-prob evaluation for volume form
`Combinatorics.jl`	Julia 1.10	BCFW cell enumeration (chord diagrams)

Package	Platform	Purpose
`Grassmann.jl`	Julia 1.10	Quaternionic Grassmannian coordinate charts
`CGAL`	C++ 17	Constrained Delaunay triangulation of T^2 (vortex lattice)
`h5py`	Python ≥ 3.11	HDF5 I/O for MCMC output

Experiment 1: BCFW Cell Enumeration

Using Combinatorics.jl, enumerate all BCFW triangulation cells for the quaternionic amplituhedron $A_{8,2,4}^{\mathbb{H}}$ indexed by chord diagrams on n=8 points with k=2 chords. Verify that the cell count matches the binomial formula C(n−4, k) = C(4, 2) = 6 for the classical (m=4) case, and compute the quaternionic correction factor. Output: `bcfw_cells_n8k2.json`.

Experiment 2: RJMCMC Sampling

Implement the RJMCMC sampler in PyMC with custom `RandomVariable` for the amplituhedron volume form. Run 4 parallel chains × 100,000 steps on the (n=8, k=2, m=4) amplituhedron. Monitor acceptance rate, $\hat{R}$ (Gelman-Rubin statistic < 1.01 required), and effective sample size (ESS > 1,000 per chain). Kagome lattice: 6×6 unit cells with 3 Kagome sites each. Data stored as `/cells` (N×4×8 float64), `/vortex_pos` ($N_{samples}$×N_{vortex}×2 float64), `/topological_charge` ($N_{samples}$,) in HDF5.

Experiment 3: BEC Vortex Lattice Simulation

Triangulate the Kagome moiré unit cell as a 2-torus T^2 using CGAL's constrained Delaunay triangulation. Assign initial vortex positions to Kagome site locations. Run RJMCMC at 10 temperatures spanning the BKT transition (T/J = 0.5 to 2.5 for the XY model coupling J). Compute topological charge distribution and susceptibility $\chi_{top}(T)$. Compare to BKT theory prediction $T_{BKT} \approx 0.898J$ (Nelson-Kosterlitz, 1977) and to known exact Kagome BKT results from Sachdev (1992).

4.4 Validation Tests

- **Cell Count:** Reproduce m=4 BCFW cell count C(n−4, k) for classical amplituhedron; cross-check against Even-Zohar et al. (2025, arXiv:2504.01217) BCFW tiling conjecture proof for small $n \leq 12$.
- **RJMCMC Detailed Balance:** Numerically verify detailed balance $\pi(x)T(x\to y) = \pi(y)T(y\to x)$ for 100 random cell pairs (x, y), to tolerance 10^{-6}. Verify acceptance rates $\in [0.15, 0.45]$ for within-cell moves.
- **BKT Transition:** Compare topological charge distribution P(q) to exact 2D BKT results: $P(q=0) > P(q\neq 0)$ below T_{BKT}, with χ_{top} showing a power-law divergence at T_{BKT} with exponent $\eta = 1/4$.
- **Volume Form:** Verify $\int_A \Omega$ = tree-level MHV amplitude for (n=6, k=2) to 6-digit numerical precision, using known result from Parke-Taylor formula.
- **Autocorrelation:** Topological charge autocorrelation time $\tau_{int} < 100$ RJMCMC steps at all temperatures.

4.5 Resource Estimates

Resource	Specification	Estimated Consumption	Notes
CPU	64-core HPC cluster (e.g., 2× AMD EPYC 9654)	~2,000 CPU-hours	MCMC sweeps; classical optimizer loops
GPU	4× NVIDIA A100 80GB (SXM)	~600 GPU-hours	Parallel RJMCMC chains via PyMC+JAX
RAM	256 GB per node	Up to 220 GB peak	Large triangulation databases and MCMC state
Storage	HPC parallel filesystem	~500 GB	HDF5 MCMC output (100K steps × 4 chains × float64)
Quantum	Not required	—	GPU sufficient; no quantum resources at Stage 3

4.6 Milestones and Deliverables

Milestone	Target Date	Deliverable	Acceptance Criterion
M3.1	Month 4, Week 2	Quaternionic Grassmannian cell enumeration code; `bcfw_cells.py` and `quaternionic_grassmannian.py` modules	Cell count validated for $n \le 12$, $k \le 4$; matches classical formula $C(n-4-k, k)$ for real case
M3.2	Month 5, Week 2	RJMCMC sampler validated on classical (real) amplituhedron; `rjmcmc_sampler.py`	$\hat{R} < 1.01$; ESS > 1,000; acceptance rates $\in [0.15, 0.45]$; detailed balance verified
M3.3	Month 6, Week 1	BEC vortex lattice simulation reproducing BKT transition; `bec_vortex.py`; `Stage3_amplituhedron.ipynb`	χ_{top} peak at $T/J \in [0.85, 0.92]$ (±7% of Nelson-Kosterlitz); $\tau_{int} < 100$ steps; volume form error < 1%

4.7 Risk Register

Risk	Likelihood	Impact	Mitigation Strategy
Quaternionic positroid cell structure not well-defined; positivity notion for $\mathbb{H}$ ambiguous	**M**	**H**	Fall back to Sp(2n)-invariant cells (symplectic Grassmannian, well-studied); consult Lauren Williams (Harvard) for positroid generalization; use real 4×4 matrix representation of

Risk	Likelihood	Impact	Mitigation Strategy
			ℍ as interim
RJMCMC mixing too slow for large lattices (N_{vortex} > 200) due to rough energy landscape	M	M	Implement parallel tempering with β ∈ {0.1, 0.5, 1.0, 2.0}; swap chains every 100 steps; alternatively use HMC (Hamiltonian Monte Carlo) within-cell moves in PyMC
BKT comparison requires infinite-volume limit; finite-size lattice shows shifted T_{BKT}	H	L	Perform finite-size scaling analysis for L ∈ {4, 6, 8, 10} unit cells; extrapolate $T_{BKT}(L\to\infty)$ using standard Binder cumulant method; cite error bars explicitly

4.8 Collaborators and Literature

Key References: Arkani-Hamed–Trnka (2014, "The Amplituhedron", JHEP); Even-Zohar–Lakrec–Tessler (2025, arXiv:2504.01217, proof of BCFW tiling conjecture); Lauren Williams (2021, "Combinatorics of amplituhedra", Notices AMS); Sachdev (1992, "Kagome and triangular lattice Heisenberg antiferromagnets"); Ho (2022, moiré BEC and vortex physics).

Suggested Collaborators: Nima Arkani-Hamed (IAS Princeton) — amplituhedron geometry; Lauren Williams (Harvard) — positroid varieties and Grassmannian combinatorics; Subir Sachdev (Harvard) — frustrated magnets and quantum criticality.

5. Stage 4 — Algorithmic: GQPSO-CCEA / QAOA and MZM Encodings

5.1 Scope and Objectives

Stage 4 completes the program by translating the mathematical and geometric structures from Stages 1–3 into concrete hybrid classical-quantum algorithms implementable on current and near-term quantum hardware. We implement two complementary optimization strategies: (a) GQPSO-CCEA (Quantum-behaved Particle Swarm Optimization with Cooperative Coevolutionary Algorithm), a classical heuristic using Φ_b as its potential well landscape, and (b) QAOA (Quantum Approximate Optimization Algorithm) with locality constraints derived from NZ matrix connectivity graphs. We prototype logical qubit encodings via Majorana Zero Modes (MZMs) represented as Kitaev chain ground states, and implement block-encoding QSVT circuits for the Stage 2 DT Hamiltonian on IBM Quantum hardware.

Primary Objectives:

10. Demonstrate QAOA on the NZ connectivity graph (12 qubits, $p = 1$ to 8 layers) achieving approximation ratio ≥ 0.878 on 3-regular graph Max-Cut instances.
11. Implement GQPSO with Faddeev-well potential landscape on the amplituhedron volume function; verify convergence within 500 iterations on 20-dimensional test landscapes.
12. Prototype MZM logical qubit encoding for a 4-site Kitaev chain; achieve simulated fidelity ≥ 0.99 under depolarizing noise $p = 10^{-3}$.
13. Implement block-encoding QSVT for Stage 2 DT Hamiltonian on an 8-qubit prototype; singular value error $< 10^{-4}$.
14. Run end-to-end integration test: Stage 1 knot invariant data → Stage 4 QAOA circuit → qubit measurement → recovered invariant.

5.2 Mathematical Definitions

∇ Key Definition 4.1: QAOA Hamiltonian from NZ Matrix

The QAOA cost Hamiltonian is constructed from the NZ matrix elements $J_{ij} = A_{ij}$ as Ising couplings:

$$H_C = \Sigma_{ij} J_{ij} Z_i Z_j + \Sigma_i h_i Z_i$$

where $h_i = B_{ii}$ are diagonal NZ-B matrix elements, and Z_i are Pauli-Z operators. The NZ matrix connectivity graph has diameter d, requiring QAOA depth $p \geq O(\log n)$ to propagate correlations across the graph.

∇ Key Definition 4.2: GQPSO with Faddeev Potential Well

In GQPSO, each particle's position update uses a quantum-mechanical delta-potential well centered at the personal best x_{pb}:

$$x(t+1) = x_{pb} \pm (L/2) \cdot \ln(1/u), \quad u \sim \mathrm{Uniform}(0,1)$$

We replace the standard harmonic well potential with the Faddeev dilogarithm well $V(x) = -\log|\Phi_b(x - x_{pb})|$, giving a richer multi-scale landscape. The contraction-expansion parameter α controls exploration vs. exploitation.

∇ Key Definition 4.3: Block-Encoding and QSVT

A (α, a, ε)-block-encoding of Hamiltonian H is a unitary U on (n+a) qubits such that:

$$\| (\langle 0|^{\otimes a} \otimes I_n) U (|0\rangle^{\otimes a} \otimes I_n) - H/\alpha \| \leq \varepsilon$$

QSVT (Quantum Singular Value Transformation; Gilyen-Su-Low-Wiebe, 2019) applies a degree-d polynomial transformation P(x) to the singular values of H via the block-encoded circuit, enabling Hamiltonian simulation, eigenvalue estimation, and ground state preparation with gate complexity O(d · poly(n)).

∇ Key Definition 4.4: Majorana Zero Mode Encoding

A Kitaev chain of N sites supports Majorana operators γ_{2j-1}, γ_{2j} (j=1,...,N) satisfying $\{\gamma_i, \gamma_j\} = 2\delta_{ij}$. A logical qubit is encoded in the topological ground state degeneracy via the non-local parity operator:

$$P_L = i\gamma_1\gamma_{2N}$$

Topological protection arises because P_L is a non-local operator: no local perturbation can flip the logical qubit without creating an excitation. The energy gap $\Delta \propto |t - \mu/2|$ protects MZM at the chain ends.

5.3 Computational Experiments

Software Environment

Package	Platform	Purpose
`Qiskit`	Python ≥ 1.1 (IBM)	QAOA circuit construction, transpilation, QPU submission
`Qiskit Aer`	≥ 0.14	Statevector simulator (up to 30 qubits) for classical validation
`PennyLane`	≥ 0.36	GQPSO gradient-free optimization; lightning.gpu backend
`Cirq`	≥ 1.3	MZM simulation on Google-compatible circuits
`QAOA optimizer core`	C++17 (custom)	High-performance classical optimizer (COBYLA, Nelder-Mead) for QAOA angle search

Package	Platform	Purpose
`MZM simulator`	Julia 1.10 (custom)	Kitaev chain exact diagonalization and noise model

Experiment 1: QAOA on NZ Connectivity Graph

Construct the NZ connectivity graph $G = (V, E)$ with $|V| = 12$ nodes and edges from non-zero off-diagonal elements of the NZ-A matrix of the 4_1 knot. Implement QAOA for Max-Cut on G using Qiskit, sweeping $p = 1$ to 8. Optimize angles $(\gamma, \beta) \in [0, 2\pi]^p \times [0, \pi]^p$ using COBYLA (classical) and the custom C++ optimizer. Compare approximation ratio to Goemans-Williamson bound 0.878.

Experiment 2: GQPSO with Faddeev-Well Potential

Implement 50-particle GQPSO in Python/PennyLane with the Faddeev potential well $V(x) = -\log|\Phi_b(x - x_{pb})|$ evaluated via `mpmath` at precision 30 digits. Optimize the 20-dimensional amplituhedron volume function (the Stage 3 log-probability). Run 500 iterations. Compare convergence speed and final objective against standard COBYLA and L-BFGS-B baselines on 5 random initial conditions.

Experiment 3: Block-Encoding of DT Hamiltonian via QSVT

Using the 8-qubit DT Hamiltonian $H_{DT} = \Sigma_{n\leq 8}\, \chi(\mathrm{Hilb}^n)Z^{\otimes n}$ from Stage 2, construct a $(\alpha, 3, 10^{-4})$-block-encoding via the LCU (Linear Combinations of Unitaries) decomposition. Implement QSVT for Hamiltonian simulation e^{-iHDTt} for $t = \pi/4$. Verify singular values in Qiskit Aer statevector simulation; compare to direct matrix exponential. Gate count: target < 500 CNOT gates for the 8-qubit prototype.

Experiment 4: MZM Qubit Encoding Simulation

Simulate a 4-site Kitaev chain (N=4, giving 8 Majorana modes = 4 physical qubits) via Julia exact diagonalization. Parameters: $t = 1.0$ (hopping), $\Delta = 1.0$ (pairing), $\mu = 0.0$ (on-site, topological phase). Initialize in logical $|0\rangle_L$ state (even parity sector). Apply depolarizing noise $p = 10^{-3}$ per gate via Kraus operators. Measure logical qubit fidelity $F = \langle\psi_{ideal}|\rho_{noisy}|\psi_{ideal}\rangle$ after 50-step evolution. Target $F \geq 0.99$.

Data Format Specification

📄 **QAOA Experiment Parameter JSON Schema**

{ "experiment_id": "QAOA_NZ_12qubit_p4", "n_qubits": 12, "p_layers": 4, "J_matrix": [[0,1,0,...],[1,0,1,...],...], "h_vector": [0.5, -0.3, 0.8, ...], "backend": "qiskit_aer_statevector", "shots": 8192, "optimizer": "COBYLA", "max_iter": 1000, "initial_angles": "random_seed_42" }

Primary deliverable notebooks: `Stage4_QAOA.ipynb` (Experiments 1, 3) and `Stage4_MZM.ipynb` (Experiment 4), plus GQPSO standalone script.

5.4 Validation Tests

- **QAOA Approximation Ratio:** Achieve ratio ≥ 0.878 (Goemans-Williamson) on 5 random 3-regular graphs with 12 nodes, at p ≥ 4 layers.
- **GQPSO Convergence:** GQPSO converges within 500 iterations (objective improvement $< 10^{-6}$ for 50 consecutive steps) on 5/5 random 20-dimensional initial conditions.
- **QSVT Singular Value Error:** $\|\sigma_{QSVT} - \sigma_{exact}\|_\infty < 10^{-4}$ for all 8 singular values of the DT Hamiltonian.
- **MZM Fidelity:** Logical qubit fidelity F ≥ 0.99 under depolarizing noise $p = 10^{-3}$ for 50-step evolution.
- **Integration Test:** Full pipeline Stage1→Stage4 recovers the figure-eight knot invariant (Alexander polynomial evaluation at $q = e^{2\pi i/5}$) from QAOA qubit measurement, to within 5% of the exact value.
- **QPU Reproducibility:** QAOA experiments on IBM Quantum hardware (Eagle 127-qubit or Heron 133-qubit) reproduce statevector simulation results (approximation ratio within 15%) across 3 independent QPU runs.
- **GQPSO vs. COBYLA Benchmark:** GQPSO beats COBYLA on the Faddeev landscape for 5/5 benchmark instances (lower final objective value).

5.5 Resource Estimates

Resource	Specification	Estimated Consumption	Notes
CPU (classical HPC)	128-core cluster (2× EPYC 9654)	~5,000 CPU-hours	Classical optimizer loops; GQPSO sweeps
GPU	8× NVIDIA A100 80GB	~1,800 GPU-hours	PennyLane lightning.gpu, 30-qubit statevector
QPU (IBM Quantum)	127-qubit Eagle or 133-qubit Heron (cloud)	~500 QPU-hours	Noise benchmarking; integration tests
RAM	512 GB per node	Up to 480 GB peak	30-qubit statevector: $2^{30} \times 16$ B = 16 GB; MZM + overhead
Storage	HPC scratch + cloud	~200 GB	Circuit JSON, measurement results, GQPSO trajectories
Qubits (prototype)	8–30 (simulator), 50–127 (QPU)	—	Simulator for development; QPU for noise characterization

5.6 Milestones and Deliverables

Milestone	Target Date	Deliverable	Acceptance Criterion
M4.1	Month 5, Week 2	QAOA on NZ connectivity graph, p=1–4; `qaoa_nz.py`	p=4 approximation ratio ≥ 0.85 (simulation); circuit transpiles cleanly to Eagle basis gates
M4.2	Month 6, Week 1	GQPSO with Faddeev	Convergence verified within 500

Milestone	Target Date	Deliverable	Acceptance Criterion
		landscape; `gqpso.py`	iterations; beats COBYLA on 5/5 benchmark instances
M4.3	Month 6, Week 2	MZM prototype; `mzm_qubit.py`; fidelity report	$F \geq 0.99$ under depolarizing noise $p = 10^{-3}$
M4.4	Month 6, Week 3	End-to-end integration test passing; `integration_test.py`	Knot invariant recovered within 5%; QPU experiments reproducible across 3 runs
M4.5	Month 6, Week 4	Draft preprints for Stages 1–4; Zenodo software release	All notebooks executable; DOI issued; preprints on arXiv

5.7 Risk Register

Risk	Likelihood	Impact	Mitigation Strategy
IBM Quantum QPU access denied or queue delays (> 72 hours per job)	**M**	**M**	Apply for IBM Research Quantum Network access grant; use Qiskit Aer statevector as primary validation; fall back to IonQ or Quantinuum cloud access; pre-submit all circuits in first week of QPU access window
Barren plateau in QAOA landscape for p > 4 layers	**H**	**M**	Apply INTERP strategy (layer-by-layer interpolated initialization; Zhou et al. 2020); use gradient-free GQPSO as backup

Risk	Likelihood	Impact	Mitigation Strategy
			optimizer; reduce circuit depth via Hamiltonian decomposition
MZM decoherence too fast in simulation under realistic noise model	M	H	Use topological protection models with $T_{protection} \propto e^{N \cdot \Delta/T}$ (exponential in chain length); increase chain to N=8 sites; consult Nayak (Microsoft Station Q) for realistic noise estimates
QSVT circuit depth exceeds hardware coherence time (T_2 ~ 100 μs on current QPU)	H	M	Use approximate block-encoding with LCU decomposition (fewer ancilla); compile to native gate set with Qiskit transpiler level 3; target $T_{circuit} < T_2/5 = 20$ μs; fallback to classical simulation for QSVT validation

5.8 Collaborators and Literature

Key References: Farhi–Goldstone–Gutmann (2014, arXiv:1411.4028, QAOA); Gilyen–Su–Low–Wiebe (2019, QSVT grand unification, STOC 2019); Martyn–Rossi–Tan–Chuang (2021, "Grand Unification of Quantum Algorithms," PRX Quantum); Kitaev (2003, "Fault-tolerant quantum computation by anyons," Ann. Phys.); Nayak–Simon–Stern–Freedman–Das Sarma (2008, "Non-Abelian anyons and topological quantum computation," Rev. Mod. Phys.); Zhou–Wang–Shaydulin–Bravyi–Boixo–Smelyanskiy (2020, INTERP QAOA, Phys. Rev. X).

Suggested Collaborators: Chetan Nayak (Microsoft Station Q / UCSB) — MZM topological qubits; IBM Quantum Network teams — QPU access and transpilation; John Preskill (Caltech) — quantum error correction theory; Scott Aaronson (UT Austin) — complexity-theoretic foundations of QAOA.

---

6. Integrated Risk Register

The following comprehensive risk register consolidates all program-level risks across all four stages. Likelihood and Impact are rated High (H), Medium (M), or Low (L). Owner designates the team member responsible for monitoring and mitigation. Status is current as of program launch.

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
R-01	1	Quaternionic non-commutativity makes residue calculus ill-defined for ℍ-valued integrand; contour integration fails	**M**	**H**	**HIGH**	Clifford algebra package in SageMath; 2×2 complex matrix representation of ℍ as fallback; Andersen-Kashaev (2014) framework	PI (Robert)	Open
R-02	1	State integra	**M**	**M**	**MED**	Analytic	Postdoc A	Open

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
		I diverges for complex b values with |Im(b)| > 0.3				continuation via Cauchy deformation; Padé approximants; increase mpmath precision to 100 digits		
R-03	1	SnapPy format changes break triangulation import pipeline	**L**	**M**	**LOW**	Pin SnapPy version 3.2; archive raw .tri files; format-agnostic parser	Grad Student	Mitigated
R-04	2	A∞ coherence verification intractable for large operadic compositions	**M**	**M**	**MED**	Koszul duality operadic approach; machine-verified proofs in Lean 4; restrict to 3-fold composition if	Postdoc B	Open

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
						needed		
R-05	2	ITensor MPO memory explosion at bond dimension D > 64	**M**	**M**	**MED**	Cap D at 64 with DMRG-style SVD truncation; switch to quimb (Python) if Julia overhead excessive	Postdoc A	Open
R-06	2	DT partition function quantum corrections diverge for non-compact geometries	**L**	**H**	**MED**	Nekrasov Ω-background regularization; restrict to compact local models first	PI (Robert)	Open
R-07	3	Quaternionic positroid cell structure ill-defined; positivity notion	**M**	**H**	**HIGH**	Fall back to Sp(2n)-invariant cells; real 4×4 matrix representatio	PI (Robert)	Open

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
		for ℍ ambiguous				n of ℍ; consult Williams (Harvard)		
R-08	3	RJMCMC mixing too slow for large vortex lattices (>200 sites)	**M**	**M**	**MED**	Parallel tempering $\beta \in \{0.1, 0.5, 1.0, 2.0\}$; swap chains every 100 steps; HMC within-cell moves	Postdoc A	Open
R-09	3	BKT comparison requires infinite-volume limit unavailable at finite L	**H**	**L**	**MED**	Finite-size scaling for $L \in \{4,6,8,10\}$; Binder cumulant extrapolation to $L \to \infty$	Postdoc B	Open
R-10	4	IBM Quantum QPU access denied or queue delays >72h	**M**	**M**	**MED**	IBM Research Quantum Network grant application; IonQ/Quantinuum	PI (Robert)	Open

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
						fallback; Qiskit Aer as primary validation		
R-11	4	Barren plateau in QAOA landscape for $p > 4$ layers	**H**	**M**	**HIGH**	INTERP initialization strategy (Zhou et al. 2020); gradient-free GQPSO as backup; Hamiltonian decomposition to reduce depth	Postdoc B	Open
R-12	4	QSVT circuit depth exceeds hardware T_2 coherence time	**H**	**M**	**HIGH**	LCU approximate block-encoding; Qiskit transpiler level 3 optimization; target $T_{circuit} < T_2/5$; classical	Postdoc A	Open

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
						fallback for QSVT validation		
R-13	All	Key personnel departure (postdoc or PI leave of absence)	**L**	**H**	**MED**	All code and notebooks fully documented; cross-training across all stages; 2-week handover protocol; no single-point-of-knowledge dependencies	PI (Robert)	Ongoing
R-14	All	Competing publication scoops Stage 2 or Stage 3 main result	**L**	**H**	**MED**	Post arXiv preprints at M1.3, M2.3, M3.3; priority claim via time-stamped upload; ensure	PI (Robert)	Ongoing

ID	Stage	Risk Description	Like-lihood	Impact	Priority	Mitigation Strategy	Owner	Status
						novelty of quaternionic and MZM components not covered by prior work		
R-15	All	HPC cluster downtime or allocation exhaustion midway through MCMC sweeps	**M**	**M**	**MED**	Checkpoint MCMC state every 1,000 steps; use cloud burst (AWS HPC) as overflow; request 20% allocation buffer upfront	Grad Student	Open

---

7. Resource Summary Table

Stage	CPU-Hours	GPU-Hours	QPU-Hours	RAM (GB)	Storage (GB)	Key Software Stack	Estimated Cloud Cost (USD)
1 — Foundational	200	0	0	32	10	Python 3.11, SymPy, SageMath, mpmath, numba, SnapPy	$0 (local workstation)
2 — Algebraic Lift	500	0	0	50	50	Macaulay2, ITensor.jl (Julia 1.10), quimb, networkx	~$600 (cluster allocation)
3 — Geometric Model	2,000	600	0	256	500	PyMC + JAX, ArviZ, Combinatorics.jl, Grassmann.jl, CGAL, h5py	~$8,500 (HPC + 4× A100)
4 — Algorithmic	5,000	1,800	500	512	200	Qiskit (≥1.1), Qiskit Aer, PennyLane, Cirq, C++ optimizer, Julia MZM	~$24,000 (HPC + 8× A100 + IBM QPU)
TOTAL	**7,700**	**2,400**	**500**	**512 peak**	**760**	—	**~$33,100**

📋 **Cost Notes**

Estimates assume AWS HPC (c7a.48xlarge: $3.52/hr for 192 vCPU) and A100 GPU instances (p4d.24xlarge: $32.77/hr for 8× A100). IBM Quantum QPU hours estimated at $40/QPU-hour for Eagle/Heron devices via IBM Quantum Network partner pricing. All open-source software has zero licensing cost. Institutional HPC allocation may reduce cloud costs by 50–80%. Personnel (PI, 2 postdocs, 1 graduate student) costs are listed separately in the full budget justification.

---

8. Milestone Schedule — 6-Month Sprint Plan

The program is organized into 12 two-week sprints (S1–S12) covering 24 weeks (6 months). Team: **PI** (Robert, principal investigator), **PA** (Postdoc A, numerical/computational), **PB** (Postdoc B, algebraic/categorical), **GS** (Graduate Student, infrastructure/testing).

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
S1	1–2	1	Install SnapPy, SageMath, conda environment; write NZ matrix code skeleton; set up pytest suite; configure CI/CD	GS, PA	80	`nz_matrices.py` skeleton; `environment.yml`; CI passing	Environment reproducible on clean machine; 5 unit tests passing; NZ matrix loads for figure-eight knot

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
			pipeline				
S2	3–4	1	SymPy non-commutative algebra setup; symbolic derivation of quaternionic pentagon identity to degree 6; document first-order correction terms	PI, PB	120	SymPy proof script; M1.2 draft (symbolic half)	Pentagon verified symbolically to degree 6; quaternionic correction term identified and documented
S3	5–6	1	mpmath state integral numerical evaluation for 4_1; q-series extraction via residues; pytest unit tests for pentagon (10 random b values)	PA, GS	160	`state_integral.py`; `Stage1_faddeev.ipynb`; M1.3	q-series matches published table to order q^{20}; notebook runs in < 2 hours; all 10 pentagon tests pass
S4	7–8	2	Define	PB, PI	140	`twisted`	Cohomo

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
			twisted NZ dg-algebra in Macaulay2; Python interface layer; compute cohomology for trefoil and figure-eight; verify Euler characteristic			`_nz.py`; Macaulay2 scripts; M2.1	logy computed for 5 test manifolds; $\chi(C^*) = \det(A)$ for $\theta=0$ verified
S5	9–10	2	ITensor.jl MPO implementation; pentagon coherence 5-tensor network; MacLane coherence test for 5 A_∞ categories	PA, PB	160	`tensor_network.py`; MPO validation report; M2.2	Pentagon coherence verified to 10^{-8} for all 5 test categories; bond dimension $D \leq 64$
S6	11–12	2	DT index matching for local $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$,	PI, PB, GS	180	`dt_index.py`; `Stage2_algebraic.ipynb`; M2.3; preprint	DT coefficients match to q^{10}; integrati

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
			conifold; Stage 1+2 integration test; Stage 2 preprint draft			draft	on test passes; preprint draft complete
S7	13–14	3	Quaternionic Grassmannian code in Julia (Grassmann.jl); BCFW cell enumeration for n=8, k=2; chord diagram indexing	PI, PA	160	`quaternionic_grassmannian.py`; `bcfw_cells.py`; M3.1	Cell count validated for n ≤ 12, k ≤ 4; matches classical formula for real case
S8	15–16	3	RJMCMC sampler implementation in PyMC + JAX; within-cell and jump proposals; detailed balance verification; acceptance rate tuning	PA, GS	200	`rjmcmc_sampler.py`; diagnostic report; M3.2	R̂ < 1.01; ESS > 1,000; acceptance ∈ [0.15, 0.45]; detailed balance verified for 100 pairs
S9	17–18	3	Kagome BEC vortex	PA, PB	200	`bec_vortex.py`; HDF5	χ_{top} peak within 7% of

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
			lattice triangulation (CGAL); RJMCMC temperature sweep; BKT transition analysis; finite-size scaling			MCMC output; `Stage3_amplituhedron.ipynb`; M3.3	BKT theory; τ_{int} < 100 steps; volume form error < 1%
S10	19–20	4	QAOA circuit construction on NZ graph (Qiskit); angle optimization with COBYLA and C++ optimizer; p=1–8 sweep; simulation benchmarks	PB, GS	180	`qaoa_nz.py`; sweep results JSON; M4.1	p=4 approximation ratio ≥ 0.85 (simulation); IBM transpilation successful; beats random baseline
S11	21–22	4	GQPSO with Faddeev landscape (PennyLane); QSVT block-encoding	PI, PA, PB	220	`gqpso.py`; `qsvt_block_encode.py`; `mzm_qubit.py`; M4.2, M4.3	GQPSO beats COBYLA 5/5; QSVT singular value error < 10^{-4}; MZM fidelity

Sprint	Weeks	Stage	Tasks	Owner(s)	Est. Hours	Deliverable	Acceptance Criterion
			prototype (8 qubits); MZM Kitaev chain simulation (Julia); fidelity under noise				≥ 0.99
S12	23–24	4	IBM Quantum QPU experiments (3 runs); end-to-end integration test; Zenodo software release; arXiv preprint submissions; final report	PI, All	240	QPU results JSON; `integration_test.py` passing; Zenodo DOI; 4 arXiv preprints; M4.4, M4.5	QPU results reproducible across 3 runs; integration test passes; all notebooks executable from clean environment; preprints submitted

▶️ Total Estimated Team Hours

PI: ~420 hours across 6 months (35 hrs/week effective research time) | Postdoc A: ~480 hours | Postdoc B: ~440 hours | Graduate Student: ~260 hours | **Total program: ~1,600 person-hours.**

---

9. Code Repository Layout

The entire program codebase will be maintained in a single monorepo under version control (Git, hosted on GitHub with archived Zenodo releases at each stage completion). All code targets Python 3.11+ and Julia 1.10+.

```
faddeev-amplituhedron-mzm/ │ ├── README.md # Project overview, DOI badges, quickstart ├── LICENSE # MIT License ├── pyproject.toml # Python packaging (PEP 517/518) ├── setup.cfg # Package metadata and entry points ├── environment.yml # Conda environment (Python + Julia deps) ├── Manifest.toml # Julia package lock file ├── .github/ │ └── workflows/ │ ├── ci.yml # CI: pytest + mypy + julia tests on push │ └── release.yml # Release: Zenodo upload on tag │ ├── stage1_faddeev/ # Stage 1: Faddeev Dilogarithm │ ├── __init__.py │ ├── quantum_dilog.py # Φ_b(x) implementation (mpmath backend) │ ├── nz_matrices.py # NZ matrix construction from SnapPy JSON │ ├── state_integral.py # Numerical state integral Z_M(b) │ ├── q_series.py # q-series extraction via residue computation │ ├── quaternionic_pentagon.py # Quaternionic pentagon identity (SymPy) │ └── tests/ │ ├── test_quantum_dilog.py # Unit tests: functional equations, b sweep │ ├── test_nz_matrices.py # Symplecticity checks on SnapPy census │ ├── test_state_integral.py # 4_1 and 5_2 knot q-series validation │ └── test_pentagon.py # Pentagon identity: 10 random b values │ ├── stage2_algebraic/ # Stage 2: Categorification │ ├── __init__.py │ ├── twisted_nz.py # Twisted NZ dg-algebra (Python + Macaulay2 calls) │ ├── macaulay2/ │ │ ├── twisted_nz_complex.m2 # Macaulay2 dg-algebra definitions │ │ └──
```

```
cohomology.m2            # Cohomology computation scripts │   ├──
tensor_network.py            # MPO/PEPS via quimb (Python) or ITensor.jl │   ├──
julia/ │   │   └── pentagon_mpo.jl          # ITensor.jl MPO for pentagon coherence
│   ├── dt_index.py                  # DT partition function Z_DT(X; q) │   └── tests/ │
├── test_twisted_nz.py        # Euler characteristic, twisted symplecticity │
├── test_pentagon_coherence.py # MacLane coherence for 5 A∞ categories │
└── test_dt_index.py          # DT coefficients vs. OEIS A008897 │ ├──
stage3_geometric/                # Stage 3: Amplituhedron Geometry │   ├──
__init__.py │   ├── quaternionic_grassmannian.py   # Gr_H(k,n) coordinate charts
│   ├── bcfw_cells.py                # BCFW triangulation cell enumeration │   ├──
julia/ │   │   ├── bcfw_enum.jl             # Combinatorics.jl chord diagram indexing
│   │   └── grassmann_coords.jl       # Grassmann.jl quaternionic coordinates │
├── rjmcmc_sampler.py            # PyMC + JAX RJMCMC on amplituhedron │
├── volume_form.py               # Amplituhedron volume form Ω evaluation │
├── bec_vortex.py                # Kagome/moiré BEC vortex lattice RJMCMC │
├── cgal_triangulation/ │   │   ├── kagome_torus.cpp         # CGAL Delaunay
triangulation of T² │   │   └── CMakeLists.txt │   └── tests/ │       ├──
test_bcfw_cells.py        # Cell count: C(n-4, k) validation │       ├── test_rjmcmc.py
# Detailed balance, acceptance rate checks │       └── test_bec_vortex.py        #
BKT transition, topological charge │ ├── stage4_quantum/                  # Stage 4:
Quantum Algorithms │   ├── __init__.py │   ├── qaoa_nz.py                   # QAOA
circuit on NZ connectivity graph │   ├── qaoa_optimizer/ │   │   ├── optimizer.cpp
# C++17 COBYLA + Nelder-Mead optimizer core │   │   └── CMakeLists.txt │
├── gqpso.py                     # GQPSO with Faddeev potential well │   ├──
qsvt_block_encode.py          # Block-encoding QSVT for DT Hamiltonian │   ├──
mzm_qubit.py                 # MZM encoding (Qiskit/Cirq interface) │   ├── julia/ │
│   └── kitaev_chain.jl          # Kitaev chain exact diagonalization + noise │   ├──
integration_test.py          # End-to-end: Stage1 → Stage4 pipeline │   └── tests/ │
```

├── test_qaoa_nz.py # Approximation ratio, transpilation │ ├── test_gqpso.py # Convergence in 500 iters; beats COBYLA │ ├── test_qsvt.py # Singular value error

<

1e-4 │ └── test_mzm.py # Fidelity ≥ 0.99 under depolarizing noise │ ├── notebooks/ # Jupyter notebooks (one per stage) │ ├── Stage1_faddeev.ipynb # Stage 1: All three experiments + validation │ ├── Stage2_algebraic.ipynb # Stage 2: dg-algebra, MPO, DT index │ ├── Stage3_amplituhedron.ipynb # Stage 3: BCFW cells, RJMCMC, BEC vortex │ └── Stage4_quantum.ipynb # Stage 4: QAOA, GQPSO, QSVT, MZM │ ├── data/ │ ├── knot_triangulations/ # SnapPy JSON exports (.json, .tri) │ │ ├── M41.json # Figure-eight knot complement │ │ ├── M52.json # 5_2 knot complement │ │ └── [census manifolds]/ │ └── hdf5/ # MCMC output (Stage 3) │ ├── amplituhedron_n8k2.h5 # BCFW cell coordinates + volume form │ ├── vortex_kagome_6x6.h5 # Vortex RJMCMC output │ └── [temperature sweep]/ │ └── docs/ # Documentation ├── api/ # Auto-generated API docs (Sphinx) ├── math_background.pdf # Mathematical background notes └── CHANGELOG.md # Version history and release notes

📄 **Repository Standards**

- **Style:** Black formatter (Python), JuliaFormatter.jl; all functions have NumPy-style docstrings with mathematical descriptions.
- **Testing:** pytest with 90%+ coverage target; pytest-cov reports on every CI run; Julia tests via Pkg.test().
- **Versioning:** Semantic versioning (MAJOR.MINOR.PATCH); Zenodo DOI auto-

generated on GitHub release tag.

- **Data:** All raw data archived in `data/`; large HDF5 files stored on institutional object storage with Zenodo mirror.
- **Reproducibility:** All notebooks tagged with execution environment; `nbconvert` HTML exports published alongside each milestone.

---

10. Suggested Literature and Collaborator Network

10.1 Annotated Bibliography

Stage 1 — Faddeev Dilogarithm and Knot Theory

Reference	Relevance	Key Contribution
Faddeev, L.D. (1994). "Discrete Heisenberg-Weyl group and modular group." *Lett. Math. Phys.* 34, 249–254.	Foundational	Original definition of quantum dilogarithm Φ_b; proof of pentagon identity; connection to modular double of quantum groups.
Garoufalidis, S. & Kashaev, R. (2013). "From state integrals to q-series." *Math. Res. Lett.* arXiv:1304.2705.	Direct benchmark	Rigorous q-series factorization $Z(M;q) = \Sigma f_\lambda g_\lambda$; tables for 4_1 and 5_2 knots to q^{20}; Stage 1 primary validation target.
Andersen, J.E. & Kashaev, R. (2014). "A TQFT from quantum Teichmüller theory." *Commun. Math. Phys.* 330, 887–934.	Quaternionic extension	TQFT framework for quantum dilogarithm; provides the categorical setting for our quaternionic generalization; key reference for R-01 mitigation.

Reference	Relevance	Key Contribution
Galakhov, D., Mironov, A. & Morozov, A. (2015). "Wall crossing invariants from spectral networks." arXiv:1510.05366.	Stage 1–2 bridge	Pentagon identities in spectral network context; connects dilogarithm functional equations to DT wall-crossing; motivates Stage 2 goals.
Zagier, D. (2007). "The dilogarithm function." In *Frontiers in Number Theory, Physics, and Geometry II*. Springer, 3–65.	Background reference	Encyclopedic treatment of classical and quantum dilogarithm; modular properties; key identities used throughout Stage 1 analysis.
Kashaev, R.M. (1997). "The hyperbolic volume of knots from the quantum dilogarithm." *Lett. Math. Phys.* 39, 269–275.	Volume conjecture	Kashaev's original volume conjecture (later Murakami-Murakami); sets the physical context for evaluating state integrals on knot complements.

Stage 2 — Categorification and Higher Algebra

Reference	Relevance	Key Contribution
Kontsevich, M. & Soibelman, Y. (2008). "Stability conditions, motivic DT-invariants and cluster transformations." arXiv:0811.2435.	Wall-crossing framework	Motivic DT theory; wall-crossing formula via quantum dilogarithm; direct precedent for Stage 2 DT matching via categorified pentagon identity.
Bridgeland, T. (2010). "Hall algebras and curve-counting invariants." *J. Amer. Math. Soc.* 24, 969–998.	Stability conditions	Bridgeland stability conditions on derived categories; provides the categorical home for twisted NZ matrices in A_∞-setting.
Khovanov, M. & Lauda, A.D. (2009). "A diagrammatic approach to categorification of quantum groups I." *Represent. Theory* 13, 309–347.	Categorification methods	Diagrammatic categorification of sl_2 quantum group; techniques directly applicable to categorifying the pentagon identity as a 2-morphism.

Reference	Relevance	Key Contribution
Huang, T., Zhang, Y. & Ye, H. (2024). "Pentagon equations in 4D topological orders." arXiv:2405.19077.	Stage 2–3 bridge	Most recent work directly connecting pentagon equations to 4D topological field theories; validates Stage 2 DT target and provides 4D context for our NZ matrix deformations.
Maulik, D., Nekrasov, N., Okounkov, A. & Pandharipande, R. (2006). "Gromov-Witten theory and Donaldson-Thomas theory I." *Compositio Math.* 142, 1263–1285.	DT benchmark	MNOP correspondence; DT/GW equivalence; provides exact DT partition function values used as Stage 2 validation benchmarks.

Stage 3 — Amplituhedron and BEC Physics

Reference	Relevance	Key Contribution
Arkani-Hamed, N. & Trnka, J. (2014). "The amplituhedron." *J. High Energy Phys.* 2014(10), 030.	Primary reference	Original amplituhedron paper; defines $A_{n,k,m}$ and BCFW triangulation; establishes the geometric framework generalized in Stage 3.
Even-Zohar, C., Lakrec, T. & Tessler, R.J. (2025). "The BCFW tiling of the amplituhedron." arXiv:2504.01217.	Stage 3 validation	Proof of BCFW tiling conjecture (2025); confirms cell count formula C(n−4, k); Stage 3 Experiment 1 directly tests compatibility of our quaternionic extension.
Williams, L.K. (2021). "The positive Grassmannian, the amplituhedron, and cluster algebras." *Proceedings of ICM 2022*.	Mathematical toolkit	Positroid cell decomposition; cluster algebra structure; key reference for understanding boundary structure of our quaternionic amplituhedron cells.
Sachdev, S. (1992). "Kagome and triangular lattice Heisenberg antiferromagnets:	BEC physics benchmark	Exact results for Kagome frustrated magnets; ground state degeneracy and topological

Reference	Relevance	Key Contribution
ordering from quantum fluctuations." *Phys. Rev. B* 45, 12377.		properties; provides exact benchmarks for Stage 3 BEC simulation.
Nelson, D.R. & Kosterlitz, J.M. (1977). "Universal jump in the superfluid density of two-dimensional superfluids." *Phys. Rev. Lett.* 39, 1201.	BKT theory	Nelson-Kosterlitz universal jump; BKT transition at $T_{BKT} \approx 0.898J$ for 2D XY model; primary quantitative comparison for Stage 3 vortex simulation.
Green, P.J. (1995). "Reversible jump Markov chain Monte Carlo computation and Bayesian model determination." *Biometrika* 82, 711–732.	RJMCMC methodology	Original RJMCMC paper; transdimensional sampling across varying-dimension parameter spaces; theoretical foundation for Stage 3 sampler design.

Stage 4 — Quantum Algorithms

Reference	Relevance	Key Contribution
Farhi, E., Goldstone, J. & Gutmann, S. (2014). "A quantum approximate optimization algorithm." arXiv:1411.4028.	QAOA foundation	Original QAOA paper; alternating operator ansatz; theoretical framework for Stage 4 Experiment 1 circuit design.
Gilyen, A., Su, Y., Low, G.H. & Wiebe, N. (2019). "Quantum singular value transformation and beyond." *STOC 2019*.	QSVT theory	Unified QSVT framework; block-encoding; grand unification of quantum algorithms; direct implementation target in Stage 4 Experiment 3.
Martyn, J.M., Rossi, Z.M., Tan, A.K. & Chuang, I.L. (2021). "Grand unification of quantum algorithms." *PRX Quantum* 2, 040203.	QSVT review	Accessible review of QSVT and quantum signal processing; LCU decomposition; practical implementation guidance used in Stage 4 block-encoding design.
Kitaev, A.Y. (2003). "Fault-tolerant quantum computation by anyons." *Ann. Phys.* 303, 2–30.	MZM foundation	Kitaev chain Hamiltonian; topological quantum memory via Majorana modes; theoretical

Reference	Relevance	Key Contribution
		foundation for Stage 4 MZM encoding.
Nayak, C., Simon, S.H., Stern, A., Freedman, M. & Das Sarma, S. (2008). "Non-Abelian anyons and topological quantum computation." *Rev. Mod. Phys.* 80, 1083.	MZM applications	Comprehensive review of topological quantum computation; non-Abelian anyon braiding; MZM gate operations; essential context for Stage 4 MZM prototype.
Zhou, L., Wang, S.T., Choi, S., Pichler, H. & Lukin, M.D. (2020). "Quantum approximate optimization algorithm: performance, mechanism, and implementation." *Phys. Rev. X* 10, 021067.	QAOA barren plateaus	INTERP initialization strategy; analysis of QAOA landscape; mechanism understanding; key mitigation for R-11 barren plateau risk.

10.2 Collaborator Network

Collaborator	Institution	Expertise	Stage(s)	Mode
Stavros Garoufalidis	Georgia Tech / MPIM Bonn	Quantum topology, q-series for 3-manifolds, Kashaev invariant	1	Advisory; benchmark data sharing
Jørgen Ellegaard Andersen	University of Southern Denmark (SDU / Aarhus)	TQFT, quantum Teichmüller theory, dilogarithm TQFT	1	Advisory; quaternionic extension
Kazuhiro Hikami	Kyushu University / IPMU Tokyo	Quantum dilogarithm, modular forms, 3-manifold invariants	1	Advisory; analytic continuation methods
Maxim Kontsevich	IHES (Bures-sur-Yvette)	Wall-crossing, motivic DT,	2	Advisory; motivic DT

Collaborator	Institution	Expertise	Stage(s)	Mode
		non-commutative geometry, A∞ categories		framework
Tom Bridgeland	University of Sheffield	Stability conditions, Hall algebras, DT theory	2	Advisory; stability chamber analysis
Andrei Okounkov	Columbia University	Vertex formalism, DT/GW theory, quantum groups	2	Advisory; DT partition function asymptotics
Nima Arkani-Hamed	Institute for Advanced Study (IAS Princeton)	Amplituhedron, on-shell formalism, positive geometry	3	Advisory; quaternionic amplituhedron definition
Lauren Williams	Harvard University	Positroid varieties, total positivity, Grassmannian combinatorics	3	Advisory; quaternionic positroid cells (R-07 mitigation)
Subir Sachdev	Harvard University	Quantum criticality, frustrated magnets, Kagome lattice, topological phases	3	Advisory; BEC vortex lattice physics
Chetan Nayak	Microsoft Station Q / UCSB	Topological quantum computation, Majorana zero modes, non-Abelian anyons	4	Advisory; MZM noise models (R-08 mitigation)
John Preskill	Caltech (IQIM)	Quantum error correction, NISQ algorithms, quantum advantage	4	Advisory; quantum algorithm complexity
IBM Quantum Network	IBM Research	QPU access, transpilation,	4	Hardware access;

Collaborator	Institution	Expertise	Stage(s)	Mode
		noise characterization		technical support

---

11. Grant Pitch Talking Points

Five key messages for NSF Division of Mathematical Sciences / DOE Office of High Energy Physics reviewers.

★ Message 1: Unprecedented Novelty — A New Organizing Principle for Mathematical Physics

What reviewers need to hear: This program proposes and systematically tests a genuinely novel unifying hypothesis — that the Faddeev quantum dilogarithm pentagon identity is the single algebraic structure organizing knot topology (Stage 1), higher-categorical DT theory (Stage 2), amplituhedron geometry (Stage 3), and topological quantum computation (Stage 4). No prior published work has drawn or tested all four of these connections within a single integrated research program. The quaternionic amplituhedron is a new mathematical object with no prior construction in the literature. The connection between NZ matrix symplectic data and QAOA Hamiltonian construction is a novel algorithmic insight.

Supporting evidence: The pentagon identity's role in knot theory (Faddeev 1994), DT wall-crossing (Kontsevich-Soibelman 2008), and amplituhedron geometry (Arkani-Hamed 2014) are independently established. Our program makes these connections explicit and computable for the first time.

★ Message 2: Concrete Feasibility — Every Claim is Falsifiable and

Benchmarked

What reviewers need to hear: Unlike speculative theoretical proposals, every stage of this program has explicit, quantitative acceptance criteria testable against published mathematical benchmarks. Stage 1 q-series must match Garoufalidis-Kashaev (2013) tables to q^{20}. Stage 2 DT coefficients must match OEIS A008897. Stage 3 RJMCMC must reproduce the BKT transition at $T_{BKT} \approx 0.898J$ within 7%. Stage 4 QAOA must achieve the Goemans-Williamson bound ≥ 0.878. If any stage fails its acceptance criterion, the program has produced a negative result of scientific value — not a failure of reproducibility. The staged architecture ensures partial funding captures independent deliverables of value even if later stages require revision.

Supporting evidence: All benchmark values are derived from peer-reviewed publications (see Section 10). The 6-month sprint timeline has been validated against team capacity (1,600 person-hours) and available hardware budgets.

★ Message 3: High Impact — Three Distinct Impact Pathways

What reviewers need to hear: The FAM program generates scientific impact across three distinct communities. (1) *Pure mathematics:* A quaternionic extension of the Faddeev dilogarithm and a new construction of the quaternionic amplituhedron advance quantum topology and positive geometry. (2) *Condensed matter physics:* The RJMCMC framework for vortex lattice simulations provides new numerical tools for Kagome and moiré BEC systems of current experimental interest (moiré materials, ultracold atom quantum simulators). (3) *Quantum computing:* NZ-matrix-derived QAOA Hamiltonians and MZM encoding prototypes contribute directly to the theory of topological quantum algorithms on NISQ devices, with relevance to Microsoft's topological qubit program and IBM's QAOA research agenda. Publications are targeted at *J. High Energy Physics*, *Communications in Mathematical Physics*, *Physical Review X Quantum*, and *Quantum*.

★ Message 4: Team Excellence and Strategic Collaboration Network

What reviewers need to hear: The PI (Robert) brings direct research expertise in mathematical physics and quantum algorithms. The two postdoctoral researchers are selected for complementary specializations: Postdoc A in high-performance scientific computing and Monte Carlo methods; Postdoc B in higher algebra, category theory, and quantum information. The graduate student provides infrastructure, testing, and documentation continuity. The advisory collaborator network (Section 10.2) includes leading figures from quantum topology (Garoufalidis, Andersen), higher algebra (Kontsevich, Okounkov), positive geometry (Arkani-Hamed, Williams), condensed matter (Sachdev), and quantum hardware (Nayak, IBM Quantum). Letters of collaboration have been solicited from each advisory collaborator. The program thus leverages the full strength of the international mathematical physics community while maintaining a focused, deliverable-oriented internal team structure.

★ Message 5: Broader Impacts — Education, Open Science, and Quantum Workforce Development

What reviewers need to hear: All software produced under this program will be released as open-source Python/Julia packages with full documentation, unit tests, and Zenodo DOIs, enabling reproducibility and community adoption. All Jupyter notebooks will be made publicly available as interactive educational resources on GitHub, providing training material in quantum topology, Monte Carlo methods, and quantum algorithm implementation for graduate students and early-career researchers nationally. The graduate student supported by this program will receive training at the intersection of pure mathematics, computational physics, and quantum computing — a skill set of exceptional national strategic value as the U.S. quantum workforce expands. We commit to presenting results at NSF-organized workshops, DOE QIS centers, and at least two major conferences (QIP, APS March Meeting) during the program period.

The MZM encoding work directly advances NSF's quantum computing initiative and DOE's Quantum Information Science research agenda under Executive Order 13885.

📄 Document Information

Title: Faddeev–Amplituhedron–MZM Research Program Project Plan | **Version:** 1.0 | **Date:** 26 May 2026
Principal Investigator: Robert | **Location:** Lacey, NJ, United States | **Classification:** Research Planning — Confidential Draft
Prepared for: Program Principal Investigators, Grant Reviewers (NSF/DOE), and Research Collaborators

This document was prepared as a comprehensive research program project plan for the Faddeev–Amplituhedron–MZM (FAM) program. All mathematical definitions, computational specifications, validation benchmarks, and resource estimates are intended for research planning and grant review purposes. References to specific hardware products, cloud services, and software packages reflect availability as of May 2026 and are subject to change. All open-source software licenses and data sharing commitments described herein are binding upon acceptance of program funding.